

Optimization of Graph Squares

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Abstract

In this paper, we investigate the matrix properties of graph squares, focusing in particular on the relationship between the adjacency matrix of a graph and the corresponding adjacency of its square.

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1 Introduction

The problem of finding the square root of a graph is a well-studied field in both mathematics and computer-science, with specific relation to distributed computing, network design and computational biology. Research in this area has primarily focused around the undirected graph H and its square $G = H^2$ which is obtained by adding an edge between every pair of vertices whose distance in H is at most two.

Much of the existing research on graph square roots has centered on characterizing when a graph admits such a root and on the complexity of reconstructing H from G . In contrast, rather than studying the construction problem directly, we analyze the relationship between the adjacency matrix A_H of the root and the adjacency matrix A_G of the square, and show that the problem can be completely categorized as an integer program.

We then examine the relaxed linear program which can be evaluated in polynomial time using interior point method, and integrate existing literature on special root structures as a stepping-stone towards generating a true polynomial time algorithm.

2 Background

Let the *graph root* $H = (V, E_H)$ be a finite, simple, undirected graph, where V is the vertex set with $|V| = n$, and $E_H \subseteq \{\{u, v\} \mid u, v \in V, u \neq v\}$ is the edge set. Denote by $d_H(u, v)$ the *graph distance* defined as the number of edges of the shortest path between vertices u and v in H . If no such path exists, $d_H(u, v) = \infty$.

The *square* of the graph H , denoted by $H^2 = (V, E_{H^2})$ is the graph on the same vertex set with edge set

$$E_{H^2} = \{uv \mid u \neq v, 1 \leq d_H(u, v) \leq 2\}. \quad (1)$$

That is, two distinct vertices are adjacent in H^2 if and only if they are connected by a path of length one or two in H . This is the standard definition of the square of a graph.

Using an alternative construction, let the *graph square* $G = (V, E_G)$ be an undirected graph defined on the same vertex set V . Let $A_H \in \{0, 1\}^{n \times n}$ denote the adjacency matrix of H , where

$$(A_H)_{ij} = \begin{cases} 1, & \text{if } ij \in E_H, \\ 0, & \text{otherwise,} \end{cases}$$

and let $D_H \in \mathbb{R}^{n \times n}$ denote the degree matrix of H , where

$$(D_H)_{ij} = \begin{cases} \deg_H(i), & \text{if } i = j, \\ 0, & \text{otherwise.} \end{cases}$$

Let $\mathbb{1}_{\geq 1} : \mathbb{R}^{n \times n} \rightarrow \{0, 1\}^{n \times n}$ denote the entrywise *indicator function*, defined by

$$\mathbb{1}_{\geq 1}(M)_{ij} = \begin{cases} 1, & \text{if } M_{ij} \geq 1, \\ 0, & \text{otherwise.} \end{cases}$$

Define the adjacency matrix of G using the adjacency and degree matrices of H

$$A_G = \mathbb{1}_{\geq 1}((A_H)^2 - D_H + A_H). \quad (2)$$

Lemma 1. *The graph G defined by (2) is equivalent to the graph square H^2 defined in (1)*

$$ij \in E_{H^2} \iff 1 \leq d_H(i, j) \leq 2. \quad (3)$$

Proof. For $i \neq j$, since D_H is diagonal, have $(D_H)_{ij} = 0$ and hence

$$((A_H)^2 - D_H + A_H)_{ij} = (A_H)_{ij}^2 + (A_H)_{ij}.$$

($\Rightarrow$) Suppose $ij \in E_G$ such that $(A_H)_{ij}^2 + (A_H)_{ij} \geq 1$. This implies that at least one of $(A_H)_{ij} = 1$ or $(A_H)_{ij}^2 \geq 1$ must be true. If $(A_H)_{ij} = 1$, then $d_H(i, j) = 1$. If $(A_H)_{ij}^2 \geq 1$, then there exists a vertex k such that $ik, kj \in E_H$, and hence $d_H(i, j) \leq 2$. Therefore $1 \leq d_H(i, j) \leq 2$, and thus $ij \in E_{H^2}$.

($\Leftarrow$) Suppose $ij \in E_{H^2}$ such that $1 \leq d_H(i, j) \leq 2$. D_H is diagonal such that $(D_H)_{ij} = 0$. If $d_H(i, j) = 1$, then $(A_H)_{ij} = 1$. If $d_H(i, j) = 2$, then there exists a vertex k such that $ik, kj \in E_H$, and hence $(A_H)_{ij}^2 \geq 1$. In either case, $(A_H)_{ij}^2 + (A_H)_{ij} \geq 1$, so $(A_G)_{ij} = 1$, and hence $ij \in E_G$.

For $i = j$, since all graphs are simple, have $(E)_{ii} = 0$, $(A)_{ii} = 0$ and hence

$$((A_H)^2 - D_H + A_H)_{ii} = (A_H)_{ii}^2 - (D_H)_{ii}.$$

$(A_H)_{ii}^2 - (D_H)_{ii} = 0$ is always true since the degree matrix of H always has diagonal values $(D_H)_{ii} = \sum_k x_{ik}$ and each edge from vertex i to its neighbour in H will always create a loop to itself in the square such that $(A_H)_{ii}^2 = \sum_{k=1} x_{ik} x_{ki} = \sum_{k=1} x_{ik}^2$ by the symmetry of adjacency matrix A_H . Since $x_{ik} \in \{0, 1\}$, the square of the boolean is identical to the boolean x_{ik} such that $(A_H)_{ii}^2 = \sum_k x_{ik} = (D_H)_{ii}$ and $(A_G)_{ii} = \mathbb{1}_{\geq 1}((A_H)^2 - D_H + A_H)$ as required.

Combining both gives that $E_G = E_{H^2}$, and so $G = H^2$. □

The formulation of the adjacency matrix of the graph square in Equation (2) is very powerful. However, the presence of the indicator function introduces a nonlinearity that prevents the problem from being expressed as a purely quadratic form, thereby limiting the direct applicability of existing results from the quadratic optimization literature.

Lemma 2. *If there are no K_3 triangle subgraphs or any C_4 square subgraphs in the graph root such that H has girth of length 5, then Equation (2) can be simplified to become*

$$A_G = ((A_H)^2 - D_H + A_H). \quad (4)$$

Proof. Since the graph root has girth of length 5 there are no cycles of length 3 or 4 in H . Thus for every pair of vertices $i \neq j \in V$ there is either no cycle that connects i, j or there is a cycle of at least length 5 between them. In all cases $(D_H)_{ij} = 0$. If there is no cycle, then there is at most one path between i, j

- i) If the path is of length 1, then $(A_H)_{ij} = 1, (A_H)_{ij}^2 = 0 \implies (A_G)_{ij} = 0 - 0 + 1 = 1$.
- ii) If the path is of length 2, then $(A_H)_{ij} = 0, (A_H)_{ij}^2 = 1 \implies (A_G)_{ij} = 1 - 0 - 0 = 1$.
- iii) If the path is of length ≥ 3 or a path does not exist, then $(A_H)_{ij} = 0, (A_H)_{ij}^2 = 0 \implies (A_G)_{ij} = 0$.

If there is a cycle of length ≥ 5 between i, j such that there are two paths between i, j

- i) If one path is of length 1, then the other is of length ≥ 4 and hence $(A_H)_{ij} = 1, (A_H)_{ij}^2 = 0 \implies (A_G)_{ij} = 0 - 0 + 1 = 1$.

ii) If one path is of length 2, then the other is of length ≥ 3 and hence $(A_H)_{ij} = 0, (A_H)_{ij}^2 = 1 \implies (A_G)_{ij} = 1 - 0 - 0 = 1$.

iii) If the paths are both of length ≥ 3 , then $(A_H)_{ij} = 0, (A_H)_{ij}^2 = 0 \implies (A_G)_{ij} = 0$.

If $i = j$, then $(A_H)_{ii} = 0, (D_H)_{ii} = \deg_H(i)$, and $(A_H)_{ii}^2 = \deg_H(i)$. Thus,

$$(A_H)_{ii}^2 - (D_H)_{ii} + (A_H)_{ii} = \deg_H(i) - \deg_H(i) + 0 = 0 = (A_G)_{ii}.$$

□

2.1 Integer Program

Let $x_{ij} \in \{0, 1\}$ be the decision variables of A_H , let $y_{ikj} \in \{0, 1\}$ be the decision variable for $x_{ik}x_{kj}$ in $(A_H)_{ij}^2$ and let $z_{ij} \in \mathbb{Z}_{\geq 0}$ represent the argument inside the indicator function.

$$D_H = \begin{bmatrix} \deg_H(1) & 0 & \cdots & 0 & 0 \\ 0 & \ddots & \vdots & \ddots & \vdots \\ \vdots & \cdots & \deg_H(i) & \cdots & 0 \\ 0 & \ddots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & \cdots & \deg_H(n) \end{bmatrix}, A_H = \begin{bmatrix} x_{11} & \cdots & x_{1j} & \cdots & x_{1n} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ x_{i1} & \cdots & x_{ij} & \cdots & x_{in} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ x_{n1} & \cdots & x_{nj} & \cdots & x_{nn} \end{bmatrix} \text{ such that}$$

$$(A_H)^2 = \begin{bmatrix} \sum_{k=1}^n x_{1k}x_{k1} & \cdots & \sum_{k=1}^n x_{1k}x_{kj} & \cdots & \sum_{k=1}^n x_{1k}x_{kn} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \sum_{k=1}^n x_{ik}x_{k1} & \cdots & \sum_{k=1}^n x_{ik}x_{kj} & \cdots & \sum_{k=1}^n x_{ik}x_{kn} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \sum_{k=1}^n x_{nk}x_{k1} & \cdots & \sum_{k=1}^n x_{nk}x_{kj} & \cdots & \sum_{k=1}^n x_{nk}x_{kn} \end{bmatrix}.$$

Then with known constants $g_{ij} \in \{0, 1\}$ from A_G , can reformulating (2) as an integer program by

equating entries of each matrix to give the following linear system of n^2 equations.

$$\begin{bmatrix} g_{11} \\ \vdots \\ g_{1j} \\ \vdots \\ g_{1n} \\ \vdots \\ g_{ij} \\ \vdots \\ g_{nn} \end{bmatrix} = \begin{bmatrix} \mathbb{1}_{\geq 1}(z_{11}) \\ \vdots \\ \mathbb{1}_{\geq 1}(z_{1j}) \\ \vdots \\ \mathbb{1}_{\geq 1}(z_{1n}) \\ \vdots \\ \mathbb{1}_{\geq 1}(z_{ij}) \\ \vdots \\ \mathbb{1}_{\geq 1}(z_{nn}) \end{bmatrix} = \begin{bmatrix} \mathbb{1}_{\geq 1}\left(\sum_{k=1}^n x_{1k}x_{k1} - (D_H)_{11} + x_{11}\right) \\ \vdots \\ \mathbb{1}_{\geq 1}\left(\sum_{k=1}^n x_{1k}x_{kj} - (D_H)_{1j} + x_{1j}\right) \\ \vdots \\ \mathbb{1}_{\geq 1}\left(\sum_{k=1}^n x_{1k}x_{kn} - (D_H)_{1n} + x_{1n}\right) \\ \vdots \\ \mathbb{1}_{\geq 1}\left(\sum_{k=1}^n x_{ik}x_{kj} - (D_H)_{ij} + x_{ij}\right) \\ \vdots \\ \mathbb{1}_{\geq 1}\left(\sum_{k=1}^n x_{nk}x_{kn} - (D_H)_{nn} + x_{nn}\right) \end{bmatrix}. \quad (5)$$

The matrix D_H is diagonal can be represented as $(D_H)_{ii} = \sum_k x_{ik}$ where $(D_H)_{ij} = 0$ for $i \neq j$. Let $M \in \mathbb{Z}_{>0}$ be a sufficiently large constant. The known z_{ij} represent the terms inside the indicator function such that

$$g_{ij} = 1 \iff z_{ij} \geq 1 \text{ and } g_{ij} = 0 \iff z_{ij} = 0.$$

The above condition can be modelled using a standard big- M formulation, and the resulting integer program is given by

Variables:

$$x_{ij} \in \mathbb{Z}_{\geq 0}, \quad \forall i, j \in \{1, \dots, n\}, \quad (6)$$

$$y_{ikj} \in \mathbb{Z}_{\geq 0}, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (7)$$

$$z_{ij} \in \mathbb{Z}_{\geq 0}, \quad \forall i, j \in \{1, \dots, n\}. \quad (8)$$

Constraints:

$$(1) \text{ Indicator argument: } z_{ii} = \sum_{k=1}^n y_{iki} - \sum_{k=1}^n x_{ik}, \forall i \in \{1, \dots, n\}, \quad (9)$$

$$z_{ij} = \sum_{k=1}^n y_{ikj} + x_{ij}, \quad \forall i \neq j \in \{1, \dots, n\}, \quad (10)$$

$$(2) \text{ Indicator function: } z_{ij} \leq M g_{ij}, \quad \forall i, j \in \{1, \dots, n\}, \quad (11)$$

$$z_{ij} \geq g_{ij}, \quad \forall i, j \in \{1, \dots, n\}, \quad (12)$$

$$(3) \text{ Product: } y_{ikj} \leq x_{ik}, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (13)$$

$$y_{ikj} \leq x_{kj}, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (14)$$

$$y_{ikj} \geq x_{ik} + x_{kj} - 1, \quad \forall i, k, j \in \{1, \dots, n\} \quad (15)$$

$$(4) \text{ Upper Bound: } x_{ij} \leq 1, \quad \forall i, j \in \{1, \dots, n\}, \quad (16)$$

$$y_{ikj} \leq 1, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (17)$$

In addition, the adjacency matrix x must satisfy further constraints arising from symmetry and diagonality. Although analogous constraints could also be imposed on the auxiliary variables z they are omitted due to redundancy and potentially introduces artificial degeneracy in the linear program.

$$(5) \text{ Symmetry: } x_{ij} = x_{ji}, \quad \forall i, j \in \{1, \dots, n\}, \quad (18)$$

$$(6) \text{ Diagonal: } x_{ii} = 0, \quad \forall i \in \{1, \dots, n\}. \quad (19)$$

Proposition 1. *Let G be a graph and consider the integer program whose variables $x_{ij} \in \{0, 1\}$ encode a graph H with adjacency matrix $A_H = (x_{ij})$. Then G has a square root H if and only if the integer program has a feasible solution x such that the corresponding graph H satisfies $H^2 = G$.*

Proof. ($\Rightarrow$) Suppose G has a square root H and let A_H be the adjacency matrix of H . $x_{ij} = (A_H)_{ij}$. Construct the variables x_{ij}, y_{ikj}, z_{ij} and the constant g_{ij} as such

i) Let $x_{ij} = 1$ if $ij \in E_H$ and $= 0$ otherwise.

ii) Let $y_{ikj} = x_{ik}x_{kj}$ for all i, k, j .

iii) Let $z_{ij} = \sum_{k=1}^n y_{ikj} + x_{ij}$ for all i, j .

iv) Let $z_{ii} = \sum_{k=1}^n y_{iki} - \sum_{k=1}^n x_{ik} = \sum_{k=1}^n x_{ik}^2 - \sum_{k=1}^n x_{ik} = 0$ since $x_{ij} \in \{0, 1\} \implies x_{ik}^2 = x_{ik}$.

v) Let $g_{ij} = 1$ if $ij \in E_G$ and $= 0$ otherwise.

Verifying constraints of the integer program for the variables

i) As $x_{ij} \in \{0, 1\} \implies y_{ikj} \in \{0, 1\}$, Equations (6)-(8) and Equations (16)-(17) are satisfied.

ii) As H is a simple directed graph, its adjacency matrix is hallow and symmetric such that Equations (18) -(19) are satisfied.

iii) As $y_{ikj} = x_{ik}x_{kj}$ and all variables are binary, Equations (13)- (15) are satisfied.

iv) Equation (10) by definition is satisfied.

v) As $z_{ii} = \sum_{k=1}^n x_{ik}x_{ki} - \sum_{k=1}^n x_{ik}$, Equation (9) is satisfied.

vi) As $z_{ij} \geq 1 \iff x_{ij} = 1$ or $\exists k$ such that $x_{ik} = x_{kj} = 1$ and as $H^2 = G$, have $g_{ij} = 1 \iff 1 \leq d_H(i, j) \leq 2$. Thus $g_{ij} = 1 \iff z_{ij} \geq 1$ so the indicator constraints Equations (11)-(12) hold.

All constraints of the integer program are satisfied by the constructed variables. Hence, the integer program is feasible.

($\Leftarrow$) Suppose the integer program on graph G has a feasible solution x , and let x_{ij} define a graph H by

$$ij \in E_H \iff x_{ij} = 1.$$

As x_{ij} satisfies the constraints of the integer program,

i) Equation (6) and Equation (16) gives $x_{ij} \in \{0, 1\}$ for all i, j ,

ii) Equation (18) and Equation (19) give $x_{ij} = x_{ji}$ and $x_{ii} = 0$,

such that x_{ij} defines a hollow, symmetric matrix with all elements being either 0 or 1 which is a valid adjacency matrix of a graph. Examining the auxiliary variables y_{ikj} and z_{ij} ,

i) Equations (13)-(15) give the strict equality $y_{ikj} = x_{ik}x_{kj}$ for all i, j, k ,

ii) Equation (10) gives $z_{ij} = \sum_{k=1}^n y_{ikj} + x_{ij} = \sum_{k=1}^n x_{ik}x_{kj} + x_{ij}$ for all $i \neq j$,

iii) Equation (9) gives $z_{ii} = \sum_{k=1}^n y_{iki} + \sum_{k=1}^n x_{ik} = \sum_{k=1}^n x_{ik}x_{ki} - \sum_{k=1}^n x_{ik} = 0$ for all i since

$$\sum_{k=1}^n x_{ik}x_{ki} = \sum_{k=1}^n x_{ik}^2 = \sum_{k=1}^n x_{ik},$$

by symmetry and since $x_{ij} \in \{0, 1\}$.

The indicator constraints Equations (11)-(12) thus give for $i \neq j$

$$g_{ij} = 1 \iff \sum_{k=1}^n x_{ik}x_{kj} + x_{ij} \geq 1 \text{ and } g_{ij} = 0 \iff \sum_{k=1}^n x_{ik}x_{kj} + x_{ij} = 0.$$

Since $x_{ij} \in \{0, 1\}$, this is equivalent to

$$g_{ij} = 1 \iff (x_{ij} = 1) \vee (\exists k \text{ such that } x_{ik} = 1 \wedge x_{kj} = 1) \text{ and } g_{ij} = 0 \iff x_{ij} = 0.$$

By definition of H this means that $g_{ij} = 1 \iff (ij \in E_H) \vee (\exists k \text{ such that } ik, kj \in E_H)$ which is equivalent to

$$g_{ij} = 1 \iff 1 \leq d_H(i, j) \leq 2.$$

Now apply Lemma 1 gives $ij \in E_G \iff ij \in E_H^2$ such that $G = H^2$. $\square$

2.2 Linear Program

The linear relaxation of the integer program defined by Equations (6)–(19) is constructed in order to generate a feasible weighted solution in polynomial time algorithm. It is done with identical constraints, however the decision variables (6), (7) and (8) are replaced with

$$x_{ij} \in \mathbb{R}_{\geq 0}, \quad \forall i, j \in \{1, \dots, n\}, \quad (20)$$

$$y_{ikj} \in \mathbb{R}_{\geq 0}, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (21)$$

$$z_{ij} \in \mathbb{R}_{\geq 0}, \quad \forall i, j \in \{1, \dots, n\}. \quad (22)$$

The results from the linear programming can give *integral solutions* and *fractional solutions* when the root exists and *fractional solutions* and *infeasible solutions* when the root does not exist such that the results in Figure 1 can be generated.

The linear relaxation potentially breaks equations in the integer program which are critical to the definition.

Consider Equations (13)-(15) which gives $y_{ikj} = x_{ik}x_{kj}$. When x is binary, this construction of y leads to y being binary as well.

Consider Equation (9), where $z_{ii} = \sum_{k=1}^n y_{iki} - \sum_{k=1}^n x_{ik} + x_{ii}$ can be simplified to $z_{ii} = \sum_{k=1}^n x_{ik}^2 - \sum_{k=1}^n x_{ik}$ and is equal to 0 when x is binary. In the linear relaxation this strict inequality no longer holds for fractional values of x_{ik} . However, when x_{ik} is fractional $x_{ik}^2 \leq x_{ik}$ such that $z_{ii} = \sum_{k=1}^n x_{ik}^2 - x_{ik} \leq 0$. As $z \in \mathbb{Z}_{\geq 0}$, this gives that $z_{ii} = 0$ as required - no additional modifications are required.

Consider Equation (10), where $z_{ij} = \sum_{k=1}^n x_{ik}x_{kj} + x_{ij}$. When $z_{ij} = 0$, the equation follows the standard definition of graph squares such that $x_{ij} = 0$ and for all intermediary ks at least one of x_{ik}, x_{kj} must be equal to 0. However, when $z_{ij} \geq 1$, fractional solutions of x_{ij} and $x_{ik}x_{kj}$ may satisfy the constraints without giving a valid integral path in the root.

3 Linear Program

3.1 Max Objective Function

Consider the addition of an objective function maximizing the total weights on all edges to the linear relaxation defined by Equations (9)–(22) in Section 2.2

$$\max \sum_{i=1}^n \sum_{j=1}^n x_{ij}. \quad (23)$$

This addition gives the improved results in Figure 2.

A maximizing objective function is considered because it pushes the linear program towards dense solutions, which are much more informative than trivial sparse solution. If a minimizing objective function is used instead, the linear program tends to avoid direct integral edges and satisfies the constraint in Equation (10) using fractional paths of length 2, distributing weight so that the summation is ≥ 1 . However, such solutions do not correspond to valid integral structures. In contrast, a maximizing objective function encourages variables to take values closer to 1, promoting direct edges and avoiding the dispersion of weight across many fractional paths of length 2. This yields more integral solutions and fewer fractional roots - see Figure 3.

Furthermore, the addition of the maximizing objective function helps with identifying square root solutions beyond those that are just unique up to isomorphism. Consider the complete graphs of which the triangle K_3 is the smallest nontrivial example. K_3 has four roots, itself, and three path graphs of length 2 that are isomorphic to each other. This ensures that the solution to the linear program remain consistent (up to isomorphism) and give useful properties for .

4 Trees

4.1 Single Triangular Decomposition

Using the local triangular structure of the graph, certain edge assignments in a square root can be fixed. For a vertex $s \in V$, define

$$T(s) = \{\{u, v\} \subseteq V \setminus \{s\} \mid (s, u) \in E, (s, v) \in E, (u, v) \in E\}$$

to be the set of 3-cliques or triangles in G that contain s . If $|T(s)| = 1$ where $T(s) = \{\{u, v\}\}$ then the vertex s exists in a unique triangle of G . When the neighbours have different degree $\deg_G(u) \neq \deg_G(v)$, then the vertex with smaller degree must be the leaf-adjacent vertex in the square root. If $\deg_G(u) = \deg_G(v)$, then either u, v can be chosen since they are connected to a clique. Importantly, if u, v is fixed by equality then it cannot be fixed again in an alternative single triangular decomposition. This is assigned deterministically using

$$x_{sa} = x_{as} = x_{ab} = x_{ba} = 1, \quad \text{where } (a, b) = \begin{cases} (u, v), & \deg_G(u) \leq \deg_G(v), \\ (v, u), & \deg_G(u) > \deg_G(v). \end{cases} \quad (24)$$

This addition gives the improved results in Figure 4 whilst maintaining polynomial runtime.

Lemma 3. *Let s, u, v form a triangle in square G where s only lies in a single clique. If $\deg_G(u) = \deg_G(v) = d \geq 2$, then*

$$N_G(u) \setminus \{s\} = N_G(v) \setminus \{s\}.$$

That is, $N_G(u) \cup \{u\}$ and $N_G(v) \cup \{v\}$ induce the same clique structure in G .

Proof. Since s lies in a single clique in G , in any tree root H the vertex s must be a leaf, and exactly one of u or v is the internal vertex adjacent to the remainder of the graph. Assume without loss of generality that u is this internal vertex. Any vertex adjacent to v in G other than s must arise from a path of length at most two through u in H , since v has no other connections in the root besides u and s . Hence

$$N_G(v) \setminus \{s\} \subseteq N_G(u) \setminus \{s\}.$$

Since $\deg_G(u) = \deg_G(v)$, both neighbourhoods have equal cardinality $|N_G(v)| = |N_G(u)|$ such that via the equality criterion for finite sets

$$N_G(v) \setminus \{s\} = N_G(u) \setminus \{s\}.$$

Hence the vertices adjacent to u and v outside of s have identical adjacency structure in G , and therefore $N_G(u) \cup \{u\}$ and $N_G(v) \cup \{v\}$ induce the same clique structure. $\square$

Proposition 2. *Let s, u, v form a triangle in square G where s only lies in a single clique. If $\deg_G(u) = \deg_G(v) = d \geq 2$, then one can choose to assign $x_{su} = x_{uv} = 1$, $x_{sv} = 0$ in the tree root H without loss of generality.*

Proof. Since s lies in a single clique in G , in the root H , s must be a leaf with either u or v being the internal vertex adjacent to the remainder of the graph. Assume without loss of generality that u is the internal vertex.

By Lemma 3, $N_G(u) \setminus \{s\} = N_G(v) \setminus \{s\}$, such that u and v induce identical clique structure in G . Therefore exchanging the roles of u and v preserves the square graph G . Hence choosing u as the internal vertex and assigning

$$x_{su} = x_{uv} = 1, \quad x_{sv} = 0$$

does not alter feasibility, since the alternative choice obtained by exchanging u and v produces an isomorphic tree root with identical square graph. $\square$

Proposition 3. *Let H be a tree and let u be an internal vertex with leaf children $s_1, \dots, s_k$ and parent v . Fixing any two leaves together with their internal vertex and parent uniquely determines the remaining leaf children of u in the linear program.*

Proof. $\square$

Ross and Harary in 1960 gave an algorithm for constructing a tree root by using characterizations of the square graph to conclude that existing tree square roots are unique up to isomorphism [RH60]. This result is strictly weaker than Ross and Harary's since any tree with an internal vertex connected only to leaves of cardinality greater than or equal to 2, apart from an edge to the remaining subgraph, will always generate a clique in the maximal root. That is, it is not a maximal tree and hence the modified linear program will not return a solution despite the input graph containing a tree root.

4.2 Tetrahedron Decomposition

A natural extension of Section 4.1 can be achieved by considering cliques of size greater than three in order to categorize tree structure whereby an internal vertex is connected to more than one leaf. However, whilst finding a clique of size $\geq k$ for fixed k is polynomial, with complexity of order $O(n^k)$ by simply checking all tuples of size k , the decision problem of whether a graph G contains a clique of size k , as well as the problem of finding a largest clique, is NP-complete. Importantly, we are only concerned with cliques of size 4, since any branching vertex with exactly two leaves will generate a 4-clique, whilst a branching vertex with more than two leaves will generate multiple 4-cliques by arbitrarily selecting any two leaves.

Consider vertices $s_1, s_2, u, v \in V$ such that $\{s_1, s_2, u, v\}$ induces a 4-clique in G . Assume

$$\deg_G(s_1) = \deg_G(s_2) = n, \quad \deg_G(s_1) < \deg_G(u) \leq \deg_G(v), \quad (25)$$

such that

$$N_G(s_1) \setminus \{s_1, s_2, u, v\} = N_G(s_2) \setminus \{s_1, s_2, u, v\} \subset N_G(u) \setminus \{s_1, s_2, u, v\} \subseteq N_G(v) \setminus \{s_1, s_2, u, v\}. \quad (26)$$

Under these neighbourhood constraints, the vertices with identical minimal neighbourhood structure correspond to leaves adjacent to a common internal vertex u , whilst the vertex v , having maximal neighbourhood containment, acts as the connecting vertex between u and the remainder of the graph.

For $n = 4$, the vertex u is a binary branching vertex contained in a unique 4-clique. In particular, when $n = 5$, the branching vertex u has three leaf children s_1, s_2, s_3 , producing $\binom{3}{2} = 3$ distinct 4-cliques

$$\{s_1, s_2, u, v\}, \quad \{s_1, s_3, u, v\}, \quad \{s_2, s_3, u, v\}.$$

More generally, suppose u has k leaf children $s_1, \dots, s_k$, and that these vertices together with v induce a $(k+2)$ -clique. Then every pair of leaves (s_i, s_j) determines a distinct 4-clique $\{s_i, s_j, u, v\}$ where the total number of such 4-cliques generated by this branching configuration is $\binom{k}{2}$. Therefore, for every 4-clique that satisfies Equation (26) can deterministically assign values to the edges

using

$$x_{s_1a} = x_{as_1} = x_{s_2a} = x_{as_2} = x_{ab} = x_{ba} = 1, \quad \text{where } (a, b) = \begin{cases} (u, v), & N_G(u) \setminus \{s_1, s_2, u, v\} \\ & \subseteq N_G(v) \setminus \{s_1, s_2, u, v\}, \\ (v, u), & N_G(v) \setminus \{s_1, s_2, u, v\} \\ & \subseteq N_G(u) \setminus \{s_1, s_2, u, v\}. \end{cases} \quad (27)$$

Proposition 4. *Let H be a tree and let u be an internal vertex with leaf children $s_1, \dots, s_k$ and parent v . Fixing any two leaves together with their internal vertex and parent uniquely determines the remaining leaf children of u in the linear program.*

Proof. Since any tree may be rooted at an arbitrary vertex, assume without loss of generality that H is rooted. Let $s_1, \dots, s_k$ be leaf children adjacent to an internal vertex u , and let v be the parent of u . Consider any pair of leaves s_i, s_j , and suppose the internal branching vertex u together with its parent v have been fixed in the linear program. Now consider any remaining leaf s_ℓ adjacent to u . Since s_ℓ is a leaf in the root, its only neighbour in the tree is u . Furthermore, all leaf children of u have identical neighbourhood structure in the square graph outside of $\{u, v, s_i, s_j\}$.

Now consider any remaining leaf s_ℓ adjacent to u . Since every leaf adjacent to u lies at distance two from s_ℓ through u , and v lies at distance two through the path $s_\ell - u - v$, the neighbourhood of s_ℓ in the square graph is

$$N_G(s_\ell) = \{u, v, s_1, \dots, s_{\ell-1}, s_{\ell+1}, \dots, s_k\}.$$

Suppose for contradiction that s_ℓ is assigned to a vertex w with $w \neq u$. Since s_ℓ is a leaf in the rooted tree, its neighbourhood in the square graph is determined by the vertices at distance at most two from w . Since $w \neq u$, the set of vertices at distance at most two from s_ℓ differs from that induced through u , giving

$$N_G(s_\ell) \neq \{u, v, s_1, \dots, s_{\ell-1}, s_{\ell+1}, \dots, s_k\}.$$

Hence either additional adjacencies are introduced into the square graph or existing adjacencies are removed and the neighbourhood of s_ℓ in the resulting square graph differs from that of the original graph, contradicting feasibility of the linear program. Thus the only feasible assignment is for s_ℓ to be adjacent to the already identified internal vertex u , implying that fixing any two leaves together with their internal vertex and parent uniquely determines all remaining leaf children adjacent to u . $\square$

4.3 Theorem

Lemma 4. *Every leaf of a tree with at least three vertices is adjacent to an internal vertex that is either adjacent to exactly one leaf or adjacent to multiple leaves.*

Proof. Let s be a leaf and let u be its unique neighbour since $\deg(s) = 1$ by definition. Since the tree has at least three vertices, u cannot be a leaf, otherwise the tree would consist only of the edge su . Hence u is an internal vertex. The only two possibilities for vertex u in relation to the leaves in its neighbourhood are

1. u is adjacent to exactly one leaf s ,
2. u is adjacent to two or more leaves including s .

□

Theorem 1. *If a square graph G has a root H that is a tree, then the single triangular decomposition and tetrahedron decomposition linear program together return an integral solution.*

Proof. Assume for contradiction that there exists a square graph G with a tree root H such that the linear program has an optimal, maximal solution x^* that is fractional. Then there exists at least one edge $ij \in E_G$ such that $0 < x_{ij}^* < 1$. Consider the root solution x^* and the following cases.

Let i be a leaf, j be its neighbour with degree 2 and let $k \neq i$ be the other neighbour of j . Let ij be the fractional edge along the path $i - j - k$ which meets the rest of the graph only at k . In G , ijk form a clique in G which by construction is also the only triangle that i can be in. As the linear program applies (24), ij pendant edge will be fixed and hence cannot be integral, hence this case cannot occur.

Let i be a leaf, j be its neighbour with degree ≥ 2 and let $k \neq i$ be another leaf of j . Let j meet the rest of the graph. Then ijk form a clique and solution x^* can be extended by simply adding an additional edge ik to form a root of larger size. This contradicts the maximality of x^* .

The remaining case is where both i and j are internal vertices. Denote the subgraph of x^* from i as H_i and the subgraph from j as H_j . Since x^* must be a tree, there are no cycles in H and hence H_i and H_j are disjoint with only one fractional bridge ij connecting them. Consider H_i, H_j and recursively apply the above logic such that we are eventually left with at least two subgraphs H_0, H_1 that do not have fractional bridges and are comprised completely of pendant vertices and leafs. From above two cases, all edges in the roots of H_0, H_1 are integral, and since the situation where a vertex has multiple leaves can not occur (there must be pendant 2-leaves together with leaves else would have above case), this completely categorizes the position of the node which connects to the rest of the graph such that there are no other automorphisms which form the same square. As this holds for both H_0 and H_1 , the endpoints of the bridge between them can be completely identified such that their bridge will be made integral. Applying this inductively thus means that H_i and H_j are completely integral such that i and j are identified and ij is made integral. This contradicts the fractional nature of x_{ij}^* .

Since no fractional, solution exists, the optimal, maximal output of the linear program must be completely integral if it results in a tree root. □

5 Symmetry

5.1 Vertex Deletion

As single triangular decomposition gives perfect characterizations of the root using the linear program for up to $n = 5$, a further refinement is obtained by analyzing fractional solutions through vertex deletion and by considering the subgraphs after this process.

Let G^- be the subgraph of G defined by $(V^- = V - \{v\}, E_{G^-} = E_G \setminus \{uv \mid u \in V^-\})$ and if it has a valid square root denote it by H^- . Let H_- be the subgraph of the root graph H satisfying the relationship $H^2 = G$ defined by $(V^-, E_H \setminus \{uv \mid u \in V^-\})$. The idea is that if G^- has a valid square root, can construct H_- from H^- by considering the following scenarios for vertex v

- i) v is connected to the subgraph V^- by only one edge to $u \in V^-$
- ii) v is connected to the subgraph V^- by two edges to $u, w \in V^-$ where $uw \in E_{H_-}$

- iii) v is connected to the subgraph G' by two edges to vertices $u, w \in G'$ where $uw \notin E_H$ and there is no path of length two between u, w in G' .
- iv) v is connected to the subgraph G' by two edges to vertices $u, w \in G'$ where $uw \notin E_H$ and there is a path of length two between u, w in G' .

where the remaining possibilities have the above sub-cases.

- i) If G^- has a valid square root H' , then $H = H' + \{v\}$ where $E_H = E'_H \cup vu$.
- ii) If G' has a valid square root H' , then $H = H' + \{v\}$ where $E_H = E'_H \cup vu$ or $E_H = E'_H \cup vw$ ($E_H = E'_H \cup vu, vw$ also satisfies but is not necessary)
- iii)

If the solution of the original linear program is fractional, starting from the vertex $v \in V$ with the smallest degree, the linear program is ran on the induced subgraph $G' = V - \{v\}$. If the resulting solution is integral, then the fractional behaviour of G is localized at v and the algorithm attempts to lift the integral structure of G' back to G by enforcing that v is attached to one of its neighbours. For each $u \in N_G(v)$, the constraint

$$x_{vu} = 1, \quad x_{uv} = 1, \quad (28)$$

is added. If the resulting solution is integral, then v is identified as a leaf vertex adjacent to u in the square root, and the solution is accepted.

If no such u yields an integral solution, the process is repeated over all vertices. This technique runs in polynomial time, as it performs at most n vertex-deletion checks and, for each such vertex v , at most $\deg_G(v) \leq n - 1$ constrained linear program solves. Since each linear program has $O(n^2)$ variables and constraints and can be solved in polynomial time $T_{LP}(n)$, the overall procedure requires at most $O(n^2 \cdot T_{LP}(n))$ time, and hence remains polynomial in n .

Proposition 5. *Vertex deletion completely categorizes all proper interval roots in polynomial time.*

Proof. □

Bipartite Symmetry

Let $A \subseteq V$ and $B \subseteq V$ be subsets of size greater than 1 being either a clique or independent with equal cardinality $|A| = |B| = m$.

Consider the case where $A \cup B = V$. Then if one of A, B is a clique and the other is independent then

Let $A \cup B = V$ where A is completely disjoint, B being a clique and each vertex $a_1, \dots, a_m \in A$ have an edge to all vertices $b_1, \dots, b_m \in B$ and vice versa. If weight of all $m(m - 1)$ edges x_{ab} in this subgraph are equal, then there exists a perfect integral matching which can be arbitrarily fixed and is assigned deterministically using

$$x_{a[0]b[0]} = x_{b[0]a[0]} = 1, \dots, x_{a[m]b[m]} = x_{b[m]a[m]} = 1, \quad (29)$$

where $a[0] < \dots < a[m], b[0] < \dots < b[m]$ are an arbitrarily but fixed lexicographical ordering of all the vertices equivalent to labelling the vertices.

Proposition 6. *If a square graph G has a maximal root H that is a split graph, then the bipartite symmetry linear program returns an integral solution.*

Proof. □

5.2 Rounding

Rounding fractional solutions such that an integral solution or the problem is deemed infeasible gives the classification in Figure 6 is obtained.

5.3 Dual Problem

We now consider the dual formulation of the max objective linear program defined in Section ?? by first restating it in canonical form.

Objective Function:

$$\min - \sum_{i=1}^n \sum_{j=1}^n x_{ij}. \quad (30)$$

Variables:

$$x_{ij} \in \mathbb{R}_{\geq 0} \left([0, 1] \right), \quad \forall i, j \in \{1, \dots, n\}, \quad (31)$$

$$y_{ikj} \in \mathbb{R}_{\geq 0} \left([0, 1] \right), \quad \forall i, k, j \in \{1, \dots, n\}, \quad (32)$$

$$z_{ij} \in \mathbb{R}_{\geq 0}, \quad \forall i, j \in \{1, \dots, n\}. \quad (33)$$

Constraints:

$$(1) \text{ Indicator argument: } z_{ii} - \sum_{k=1}^n y_{iki} + \sum_{k=1}^n x_{ik} - x_{ii} = 0, \forall i \in \{1, \dots, n\}, \quad (34)$$

$$z_{ij} - \sum_{k=1}^n y_{ikj} - x_{ij} = 0, \quad \forall i \neq j \in \{1, \dots, n\}, \quad (35)$$

$$(2) \text{ Indicator function: } z_{ij} - M g_{ij} \leq 0, \quad \forall i, j \in \{1, \dots, n\}, \quad (36)$$

$$-z_{ij} + g_{ij} \leq 0, \quad \forall i, j \in \{1, \dots, n\}, \quad (37)$$

$$(3) \text{ Product: } y_{ikj} - x_{ik} \leq 0, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (38)$$

$$y_{ikj} - x_{kj} \leq 0, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (39)$$

$$x_{ik} + x_{kj} - y_{ikj} - 1 \leq 0, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (40)$$

$$(4) \text{ Symmetry: } x_{ij} - x_{ji} = 0, \quad \forall i, j \in \{1, \dots, n\}, \quad (41)$$

$$(5) \text{ Diagonal: } x_{ii} = 0, \quad \forall i \in \{1, \dots, n\}, \quad (42)$$

$$(6) \text{ Variable Bound: } x_{ij} - 1 \leq 0, \quad \forall i, j \in \{1, \dots, n\}, \quad (43)$$

$$y_{ikj} - 1 \leq 0, \quad \forall i, k, j \in \{1, \dots, n\}. \quad (44)$$

This gives the dual problem with

Objective Function:

$$\max \sum_{i=1}^n \sum_{j=1}^n \left(-M g_{ij} \gamma_{ij} + g_{ij} \delta_{ij} - \lambda_{ij} \right) - \sum_{i=1}^n \sum_{k=1}^n \sum_{j=1}^n (\eta_{ikj} + \mu_{ikj}). \quad (45)$$

Dual Variables:

$$\alpha_i \in \mathbb{R}, \quad \forall i \in \{1, \dots, n\}, \quad (\text{associated with (34)}), \quad (46)$$

$$\beta_{ij} \in \mathbb{R}, \quad \forall i \neq j \in \{1, \dots, n\}, \quad (\text{associated with (35)}), \quad (47)$$

$$\gamma_{ij} \geq 0, \quad \forall i, j \in \{1, \dots, n\}, \quad (\text{associated with (36)}), \quad (48)$$

$$\delta_{ij} \geq 0, \quad \forall i, j \in \{1, \dots, n\}, \quad (\text{associated with (37)}), \quad (49)$$

$$\epsilon_{ikj} \geq 0, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (\text{associated with (38)}), \quad (50)$$

$$\zeta_{ikj} \geq 0, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (\text{associated with (39)}), \quad (51)$$

$$\eta_{ikj} \geq 0, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (\text{associated with (40)}), \quad (52)$$

$$\theta_{ij} \in \mathbb{R}, \quad \forall i, j \in \{1, \dots, n\}, \quad (\text{associated with (41)}), \quad (53)$$

$$\iota_i \in \mathbb{R}, \quad \forall i \in \{1, \dots, n\}, \quad (\text{associated with (42)}), \quad (54)$$

$$\lambda_{ij} \geq 0, \quad \forall i, j \in \{1, \dots, n\}, \quad (\text{associated with (43)}), \quad (55)$$

$$\mu_{ikj} \geq 0, \quad \forall i, k, j \in \{1, \dots, n\}, \quad (\text{associated with (44)}). \quad (56)$$

Dual Constraints:

$\forall i, j \in \{1, \dots, n\}$, each primal x_{ij} :

$$\begin{cases} -1 - \sum_{k=1}^n \epsilon_{iik} - \sum_{k=1}^n \zeta_{kii} + \sum_{k=1}^n \eta_{iik} + \sum_{k=1}^n \eta_{kii} + \iota_i + \lambda_{ii} \geq 0, & \text{if } j = i, \\ -1 + \alpha_i - \beta_{ij} - \sum_{k=1}^n \epsilon_{ikj} - \sum_{k=1}^n \zeta_{kij} + \sum_{k=1}^n \eta_{ikj} + \sum_{k=1}^n \eta_{kij} + \theta_{ij} - \theta_{ji} + \lambda_{ij} \geq 0, & \text{if } j \neq i, \end{cases} \quad (57)$$

$\forall i, k, j \in \{1, \dots, n\}$, each primal y_{ikj} :

$$\begin{cases} -\alpha_i + \epsilon_{iki} + \zeta_{iki} - \eta_{iki} + \mu_{iki} \geq 0, & \text{if } j = i, \\ -\beta_{ij} + \epsilon_{ikj} + \zeta_{ikj} - \eta_{ikj} + \mu_{ikj} \geq 0, & \text{if } j \neq i, \end{cases} \quad (58)$$

$\forall i, j \in \{1, \dots, n\}$, each primal z_{ij} :

$$\begin{cases} \alpha_i + \gamma_{ii} - \delta_{ii} = 0, & \text{if } j = i, \\ \beta_{ij} + \gamma_{ij} - \delta_{ij} = 0, & \text{if } j \neq i. \end{cases} \quad (59)$$

whereby the Lagrangian is

$$\begin{aligned} \mathcal{L}(x, y, z) = & - \sum_{i=1}^n \sum_{j=1}^n x_{ij} + \sum_{i=1}^n \alpha_i \left(z_{ii} - \sum_{k=1}^n y_{iki} + \sum_{k=1}^n x_{ik} - x_{ii} \right) + \sum_{i \neq j} \beta_{ij} \left(z_{ij} - \sum_{k=1}^n y_{ikj} - x_{ij} \right) \\ & + \sum_{i=1}^n \sum_{j=1}^n \gamma_{ij} (z_{ij} - M g_{ij}) + \sum_{i=1}^n \sum_{j=1}^n \delta_{ij} (-z_{ij} + g_{ij}) + \sum_{i=1}^n \sum_{k=1}^n \sum_{j=1}^n \epsilon_{ikj} (y_{ikj} - x_{ik}) \\ & + \sum_{i=1}^n \sum_{k=1}^n \sum_{j=1}^n \zeta_{ikj} (y_{ikj} - x_{kj}) + \sum_{i=1}^n \sum_{k=1}^n \sum_{j=1}^n \eta_{ikj} (x_{ik} + x_{kj} - y_{ikj} - 1) \\ & + \sum_{i=1}^n \sum_{j=1}^n \theta_{ij} (x_{ij} - x_{ji}) + \sum_{i=1}^n \iota_i x_{ii} + \sum_{i=1}^n \sum_{j=1}^n \lambda_{ij} (x_{ij} - 1) + \sum_{i=1}^n \sum_{k=1}^n \sum_{j=1}^n \mu_{ikj} (y_{ikj} - 1). \end{aligned}$$

Collecting terms gives

$$\begin{aligned}
\mathcal{L}(x, y, z) = & \sum_{i=1}^n \sum_{j=1}^n x_{ij} \left[-1 + \mathbf{1}_{\{j=i\}} \left(\alpha_i - \sum_{k=1}^n \epsilon_{iki} - \sum_{k=1}^n \zeta_{kii} + \sum_{k=1}^n \eta_{iki} + \sum_{k=1}^n \eta_{kii} + \iota_i + \lambda_{ii} \right) \right. \\
& \left. + \mathbf{1}_{\{j \neq i\}} \left(\alpha_i - \beta_{ij} - \sum_{k=1}^n \epsilon_{ikj} - \sum_{k=1}^n \zeta_{kij} + \sum_{k=1}^n \eta_{ikj} + \sum_{k=1}^n \eta_{kij} + \lambda_{ij} \right) + \theta_{ij} - \theta_{ji} \right] \\
& + \sum_{i=1}^n \sum_{k=1}^n \sum_{j=1}^n y_{ikj} \left[-\mathbf{1}_{\{j=i\}} \alpha_i - \mathbf{1}_{\{j \neq i\}} \beta_{ij} + \epsilon_{ikj} + \zeta_{ikj} - \eta_{ikj} + \mu_{ikj} \right] \\
& + \sum_{i=1}^n \sum_{j=1}^n z_{ij} \left[\mathbf{1}_{\{j=i\}} \alpha_i + \mathbf{1}_{\{j \neq i\}} \beta_{ij} + \gamma_{ij} - \delta_{ij} \right] \\
& - \sum_{i=1}^n \sum_{j=1}^n M g_{ij} \gamma_{ij} + \sum_{i=1}^n \sum_{j=1}^n g_{ij} \delta_{ij} - \sum_{i=1}^n \sum_{k=1}^n \sum_{j=1}^n \eta_{ikj} - \sum_{i=1}^n \sum_{j=1}^n \lambda_{ij} - \sum_{i=1}^n \sum_{k=1}^n \sum_{j=1}^n \mu_{ikj}.
\end{aligned}$$

This dual formulation

References

- [RH60] Ian C. Ross and Frank Harary. “The square of a tree”. In: *The Bell System Technical Journal* 39.3 (1960), pp. 641–647.

A Result Diagrams

A.1 Linear Program

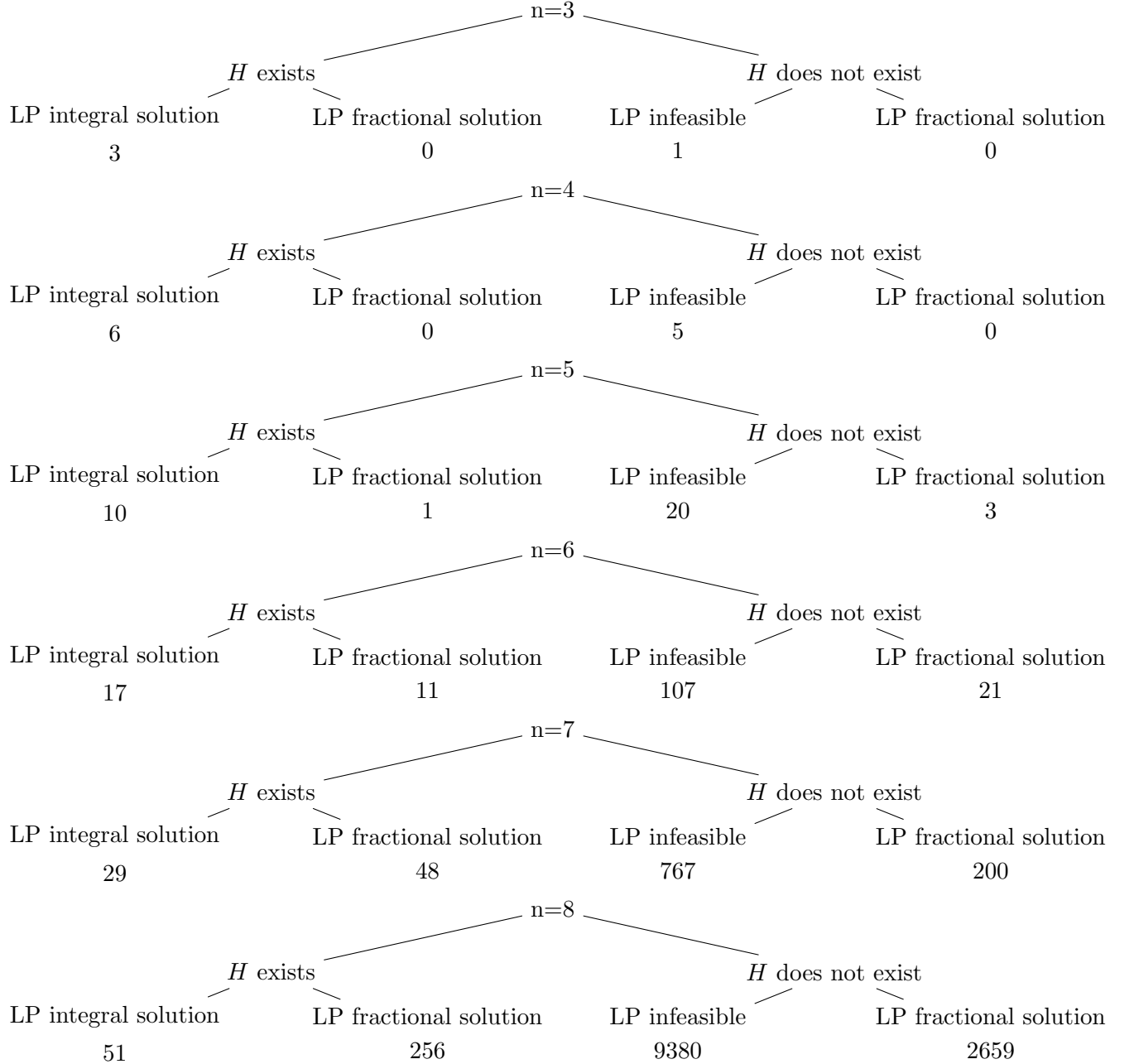


Figure 1: LP outcomes for different values of n on base linear program.

1

¹The total number of simple graphs on $n = 2, 3, 4, \dots$ unlabeled nodes is the sequence 2, 4, 11, 34, 156, 1044, 12346, $\dots$ given by A000088.

The IP integral solution sequence 2, 3, 6, 10, 17, 29, 51, $\dots$ is given by A285665.

The total number of graphs which are squares is the sequence 2, 3, 6, 11, 28, 77, 307, $\dots$ given by A382181.

A.1.1 Max Objective Function

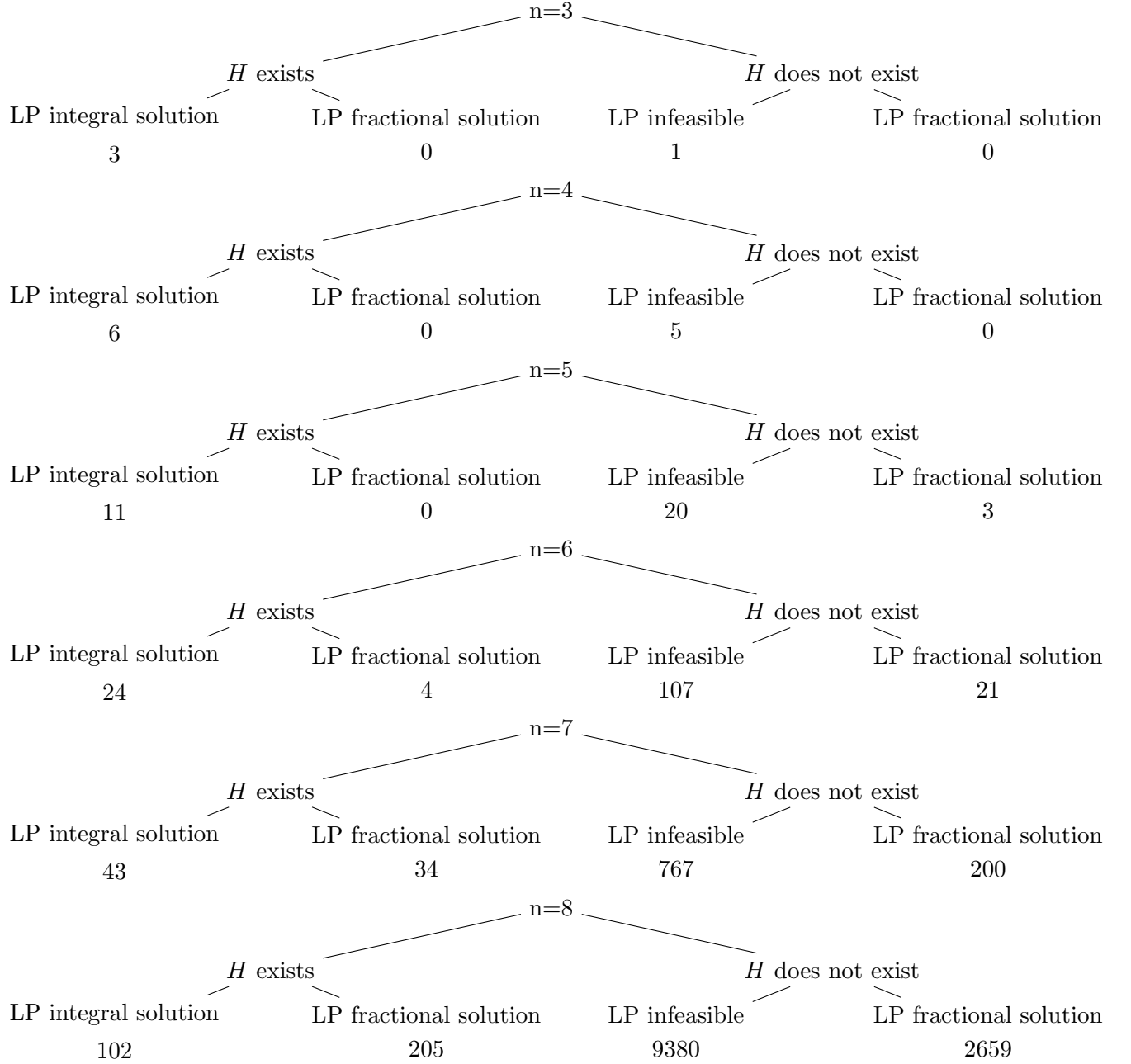


Figure 2: LP outcomes for different values of n on the linear program with max objective function.

2

²The LP integer solution sequence 3, 6, 11, 22, 42, 102, ... is very similar to the number of distinct characteristic polynomials of trees with n nodes - A226594.

A.1.2 Min Objective Function

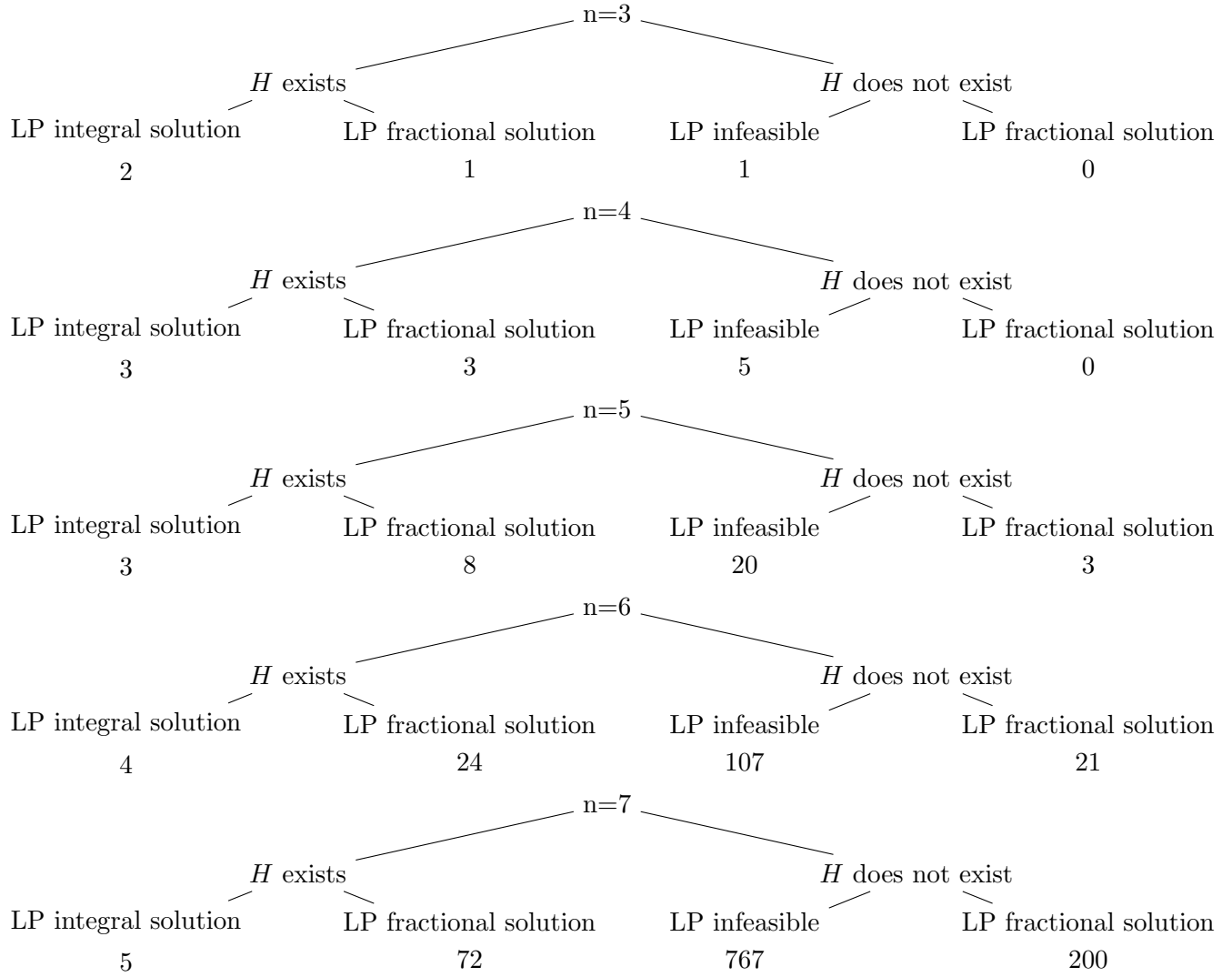


Figure 3: LP outcomes for different values of n on the linear program with min objective function.

A.1.3 Single Triangular Decomposition

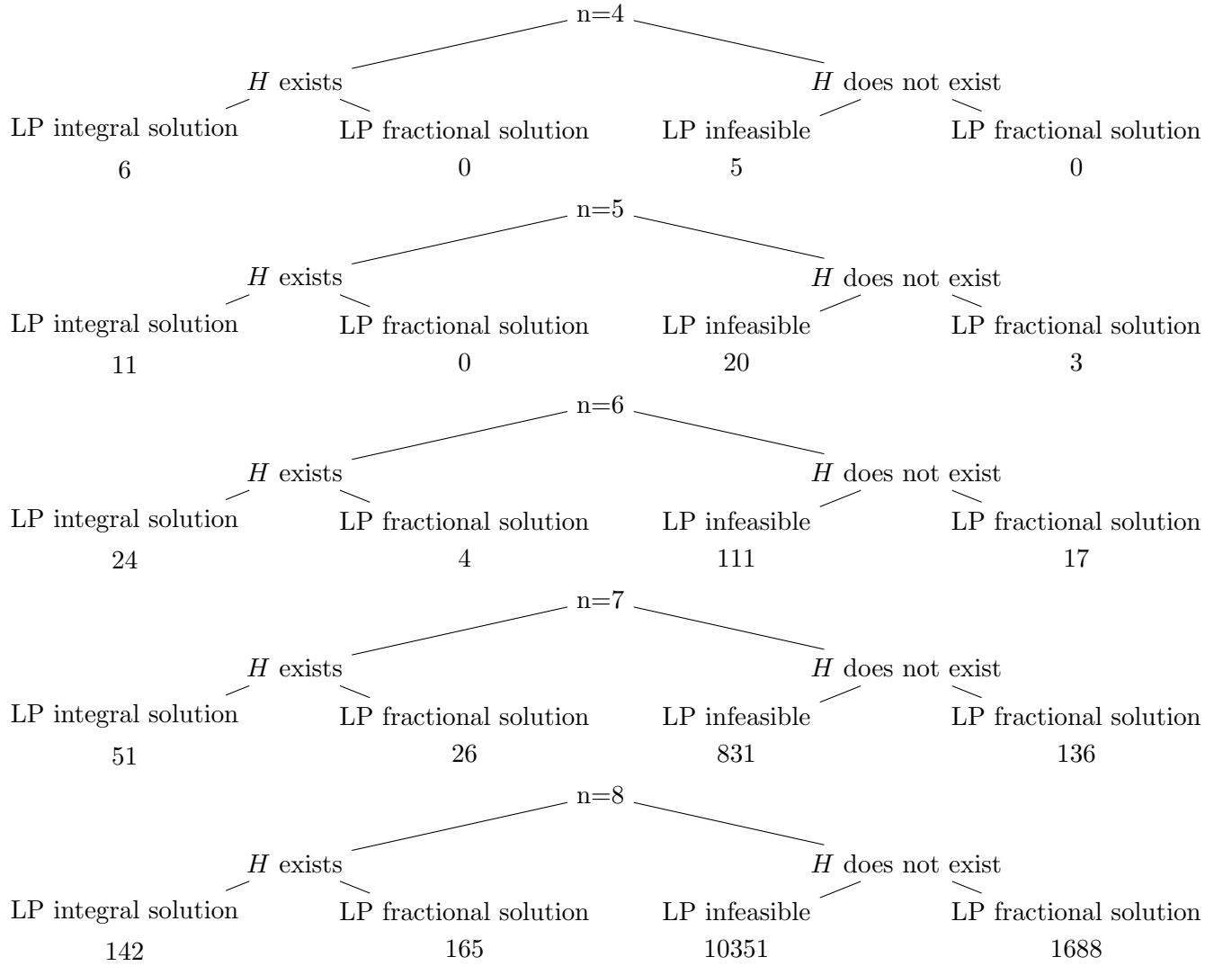


Figure 4: LP outcomes for different values of n on the linear program with max objective function and single triangular decomposition.

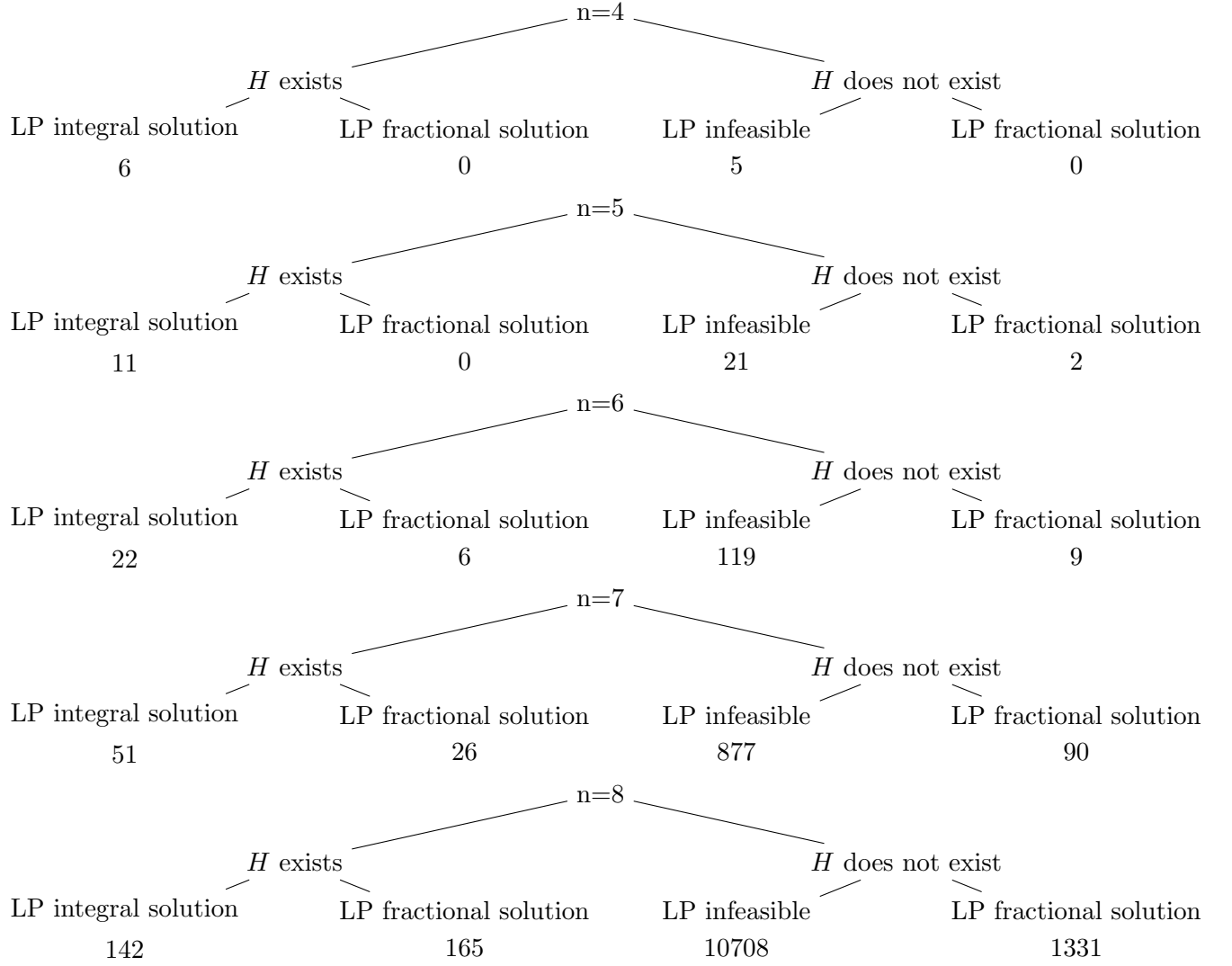


Figure 5: LP outcomes for different values of n on the linear program with max objective function and single triangular decomposition with recursion for tie-breaking.

A.1.4 Rounding

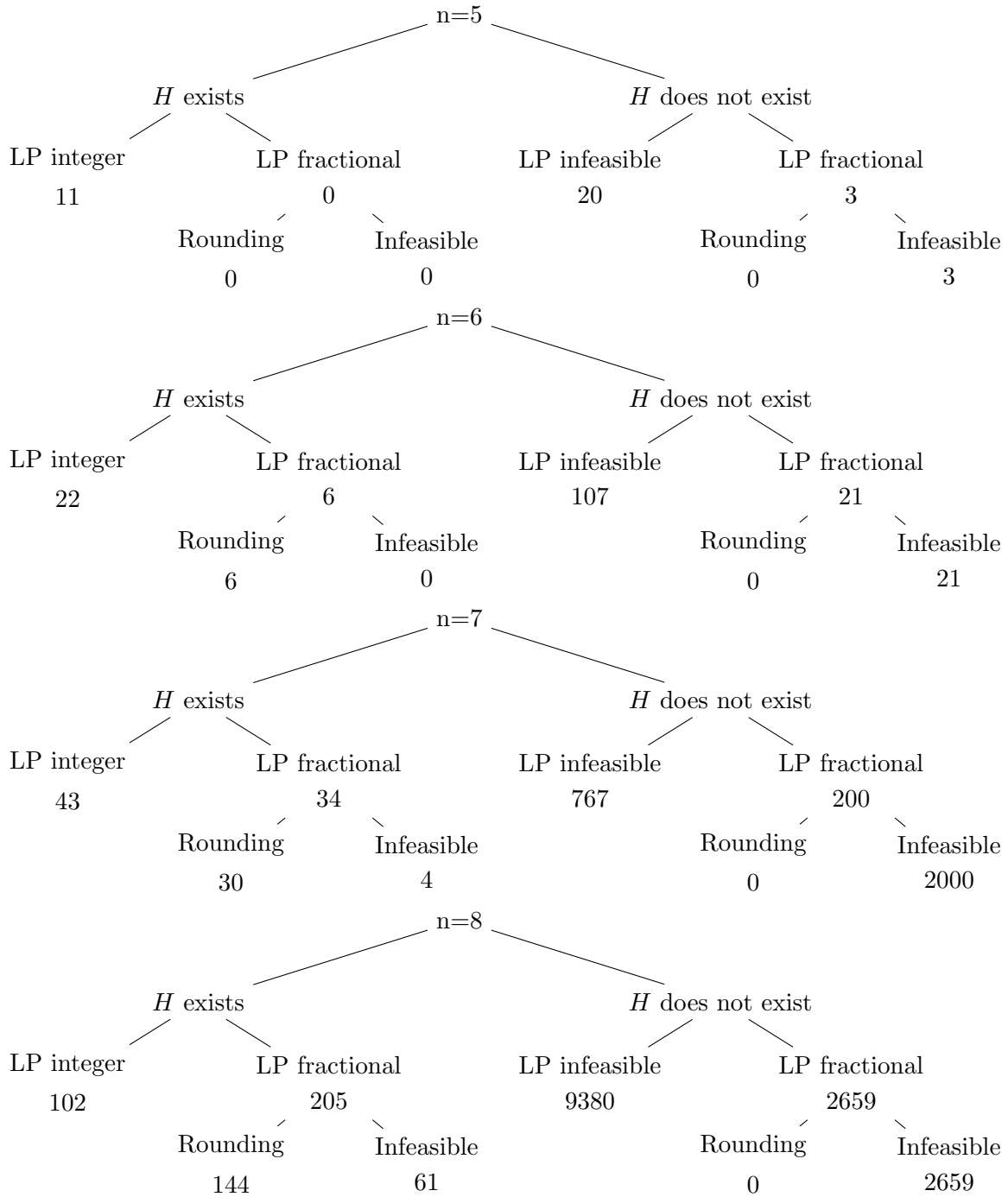


Figure 6: LP outcomes for different values of n on the linear program with max objective function and using rounding for fractional solutions.

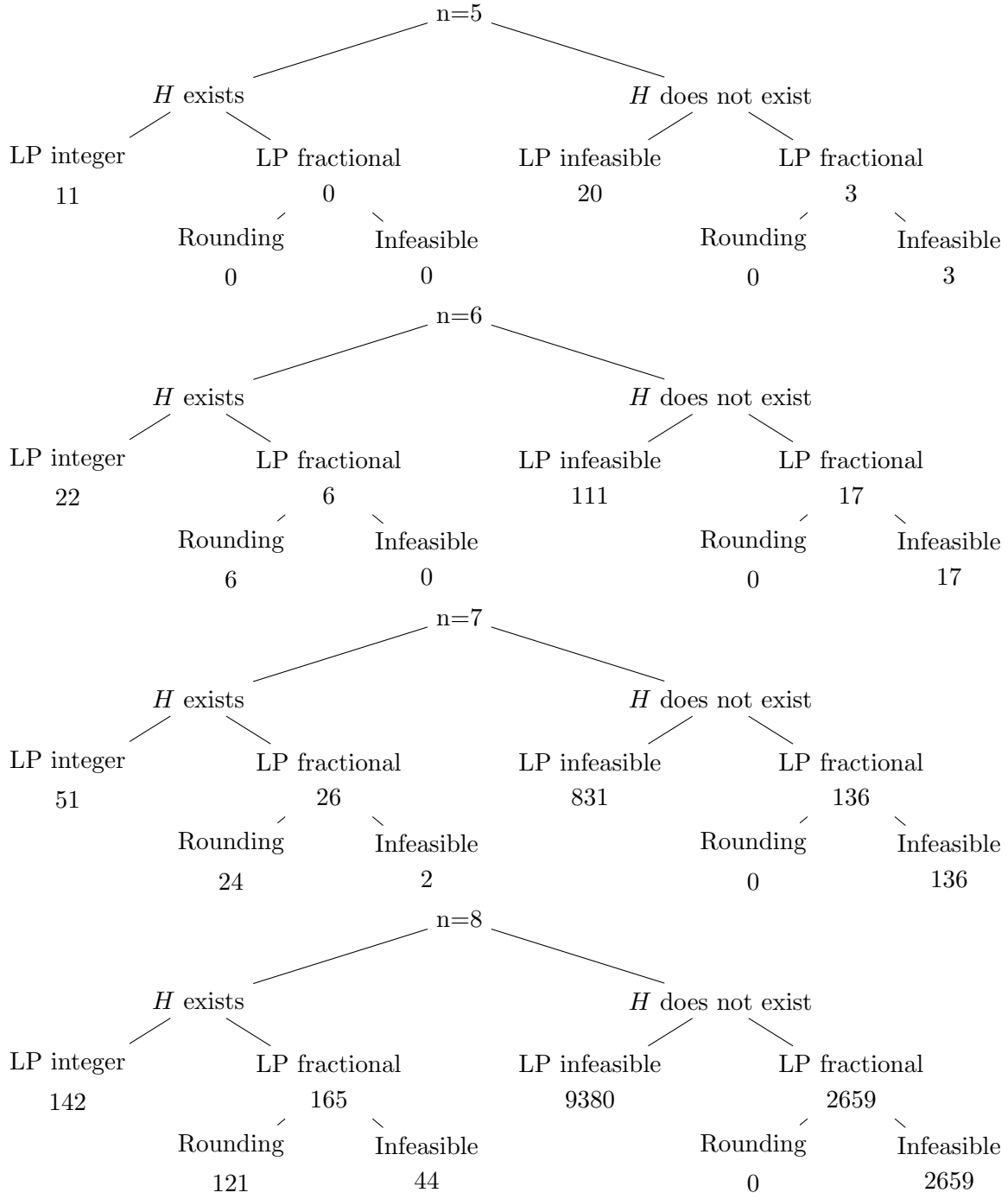


Figure 7: LP outcomes for different values of n on the linear program with max objective function, single triangular decomposition and using rounding for fractional solutions.

A.2 Dual Program

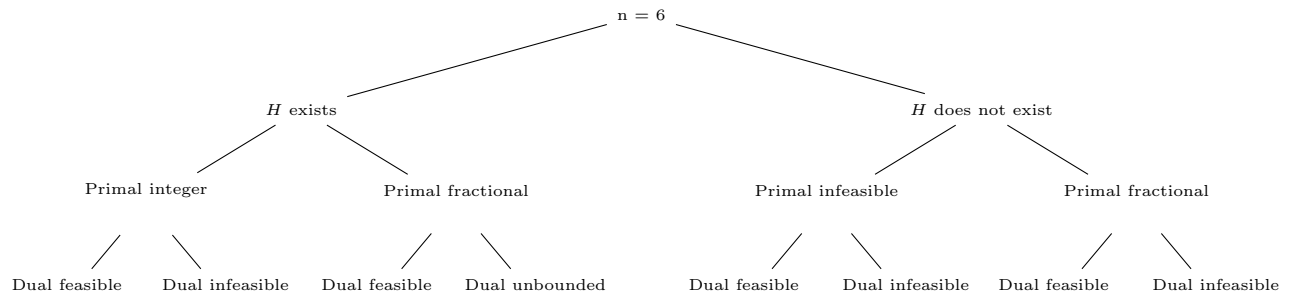


Figure 8: LP outcomes for different values of n on the dual linear program.